

TypeScript

JavaScript with types · Equigrow short notes

Why

TypeScript adds types to JavaScript so mistakes are caught while you type, not when users click. It compiles to plain JavaScript.

Basics

```
let count: number = 0;
const name: string = "Asha";
function add(a: number, b: number): number { return a + b; }
interface User { id: number; name: string; email?: string } // ? = optional
const u: User = { id: 1, name: "Asha" };
type Status = "todo" | "done"; // union type
```

Useful

- Arrays: `number[]` or `Array<number>`. Generics: `function first<T>(a: T[]): T`.
- Avoid **any**; use **unknown** when the type is truly unknown.
- Start a project: `npm create vite@latest -- --template react-ts`.